A CENTRALIZED SYSTEM FOR REAL-TIME FARM ANIMAL DISEASE REPORTING

INDIVIDUAL THESIS

W.M.P.J Wijenayake

IT22194558

Individual component: Persistent Health Anomaly & Body Condition Scoring (BCS) Module

Group ID: R26-SE-013

Report submitted in partial fulfilment of the requirements for the

Bachelor of Science (Hons) in Information Technology

Specialized in Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. I declare that this thesis is my own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. I grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this thesis, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

Academic integrity and AI disclosure. I have disclosed all use of generative AI and other automated tools in accordance with the module policy. I remain responsible for the accuracy, originality, citations, code, data, analyses, and conclusions in this thesis.

Student name: W.M.P.J Wijenayake

Student ID: IT22194558

Signature: _____ Date: _____

Supervisor certification. The above candidate carried out this research for the bachelor's degree thesis under my supervision.

Role and name	Signature	Date
Supervisor: Ms. Dilani Lunugalage		
Co-supervisor: Mr. Eishan Weerasinghe		

ABSTRACT

Reactive veterinary treatments lead to significant economic losses, since therapeutic measures are often not taken until animals are clinically affected in tropical livestock production systems. Even if continuous monitoring infrastructure and expensive 3D sensors could be installed, the cost remains too high for smallholder farmers, thus making Precision Livestock Farming (PLF) economically unviable for small farms. Furthermore, current systems are unable to correlate visual physiological data with environmental stressors, resulting in a significant diagnostic triage gap. This research aims to solve this by developing an AI-driven, on-demand, proactive standardization module for wellness triage and management. The adopted solution consists of a two-stage computer vision pipeline: (1) dynamic localization of the bovines using YOLOv8s, and (2) BCS estimation from standard 2D mobile images, utilizing a custom Convolutional Neural Network (CNN) trained for continuous BCS regression. In addition, a Late-Fusion Neural Network combines three modalities—a visual BCS, a 7-day time-series of physiological data processed through an LSTM branch, and static genetic metadata—alongside real-time GPS-derived Temperature-Humidity Index (THI) data. This multi-modal architecture is able to correctly classify the state of the animal's well-being as Healthy, Heat Stress, or Unhealthy. One of the key novelties of this study is the development of a regression-specific Gradient-weighted Class Activation Mapping (Grad-CAM) approach that avoids gradient collapse by computing absolute gradients to produce highly accurate anatomical heatmaps and an automatic targeting crosshair for vets. Experimental validation reveals that the multimodal late-fusion approach with dynamic thermal context is able to detect pre-clinical physiological deviations that are not captured by unimodal baselines. Overall, the system successfully fills the triage gap by delivering an accessible, low-cost, and transparent decision-support system for smallholder farmers to help them manage thermal stress, implement early interventions, and consolidate livestock health management.

Keywords: Proactive Triage, Multi-modal Late-Fusion, Body Condition Scoring (BCS), Explainable AI (XAI), Temperature-Humidity Index (THI)

ACKNOWLEDGEMENTS

I would like to thank my research supervisor Ms. Dilani Lunugalage and co-supervisor Mr. Eishan Weerasinghe for their guidance, encouragement and constructive criticism during this research. They brought a wealth of knowledge and willingness to help in the technical design and clinical validity of this study.

I also gratefully acknowledge the assistance of the academic and administrative staff of the Department of Information Technology, Sri Lanka Institute of Information Technology (SLIIT), in providing the necessary technical and research facilities.

I would like to specially thank the members of my research team, A.L.M. Athulathmudali, G.A. Sandaru and S.S. Kumarasinghe. Their teamwork, many hours of hard work, and vision were vital in creating the overall ADRS (Automated Cattle Wellness Monitoring System) framework.

In addition, I am very thankful to the domain experts from the Department of Animal Production and Health (DAPH), field veterinarians, and local dairy farmers. The fact that they were willing to share their practical experience, help collect primary 2D image data, and annotate the data sets for the ground truth, made this research more applicable to the real world.

Last but not least, I would like to thank my family, particularly my father, for their continual encouragement and support that has been constant and never-ending throughout my academic journey. I would not have reached this milestone without his unwavering confidence in me.

TABLE OF CONTENTS

1 INTRODUCTION 1

1.1 Overall project context 1

1.2 Individual component context 1

1.3 Problem statement 2

1.4 Research gap 2

1.5 Aim and objectives 3

1.6 Research questions or hypotheses 4

1.7 Scope limitations and assumptions 4

1.8 Individual contribution 5

1.9 Thesis structure 5

2 LITERATURE REVIEW AND RELATED WORK 7

2.1 Review method and scope 7

2.2 Domain and theoretical foundations 7

2.3 Related research 8

2.4 Existing systems products and techniques 9

2.5 Gap synthesis and conceptual framework 10

3 RESEARCH METHODOLOGY 12

3.1 Research design 12

3.2 Data participants or study materials 13

3.3 Experimental or study procedure 14

3.4 Measures and evaluation metrics 15

3.5 Data analysis 15

3.6 Ethics privacy security and responsible AI 16

3.7 Reproducibility and research artefacts 17

4 PRODUCT AND SOLUTION DESIGN 18
4.1 Requirements 18
4.2 Component architecture within the complete solution 20
4.3 Design alternatives and decisions 21
4.4 Detailed design 23
4.5 Standards and professional practice 28
5 IMPLEMENTATION INTEGRATION AND TESTING 30
5.1 Development environment and technology stack 30
5.2 Component implementation 31
5.3 Integration with the other components 31
5.4 Verification and software testing 32
5.5 Deployment and operation 33
6 RESULTS AND VALIDATION 34
6.1 Research results 34
6.2 Product and technical results 36
6.3 User or stakeholder validation 37
6.4 Integration results 37
6.5 Summary of findings 38
7 DISCUSSION 39
7.1 Interpretation of findings 39
7.2 Comparison with literature baselines and existing solutions 39
7.3 Research contribution 40
7.4 Product contribution 40
7.5 Limitations and threats to validity 40
7.6 Implications 41
8 CONCLUSION AND FUTURE DIRECTIONS 42

8.1 Conclusion..... 42
8.2 Achievement of objectives..... 42
8.3 Individual research and product contribution..... 43
8.4 Future directions..... 43
REFERENCES..... 45
GLOSSARY..... 46
APPENDICES..... 48
Appendix B Academic integrity and AI use disclosure..... 48

LIST OF FIGURES

Figure 1 High-Level System Architecture Diagram 21
Figure 2:Wellness Data Intake with Alert..... 24
Figure 3:7-Day Wellness Triage Intake 25
Figure 4:Diagnostic Results displaying the classification, calculated BCS, and the Grad-CAM XAI heatmap..... 26
Figure 5:Actionable Management Protocols..... 27
Figure 6:The Standalone BCS Analyzer visualizing YOLOv8 object detection and CNN ROI extraction. 28
Figure 7:Training evaluation output of the optimized Random Forest Persistent Monitoring Model achieving 99.80% accuracy. 34
Figure 8:Comparative evaluation demonstrating the 100.0% accuracy of the optimized Base Model versus the overfitting Expanded Model. 35
Figure 9 : The Diagnostic Triage Results displaying the classification, calculated BCS, and the highly interpretable Grad-CAM XAI heatmap. 37

LIST OF TABLES

Table 1 : Functional Requirements for the ADRS Triage Component 18
Table 2 : Evaluation of Technology Stack Alternatives 22

LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADRS	Automated Cattle Wellness Monitoring System
AI	Artificial Intelligence
API	Application Programming Interface
BCS	Body Condition Score
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
DAPH	Department of Animal Production and Health
DSR	Design Science Research
GPS	Global Positioning System
Grad-CAM	Gradient-weighted Class Activation Mapping
IOT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
LSTM	Long Short-Term Memory
PLF	Precision Livestock Farming
SDG	Sustainable Development Goal
THI	Temperature-Humidity Index
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall project context

In Sri Lanka, the agricultural sector, particularly the dairy industry, plays a pivotal role in the country's economy and food security [1]. In practice, however, the industry faces substantial economic losses due to its reactive healthcare strategy, whereby veterinary care and interventions are applied only after severe clinical symptoms are observed [2]. To address this challenge, the overarching Automated Cattle Wellness Monitoring System (ADRS) was proposed. The ADRS is a proactive, four-tiered intelligence framework designed for tropical small-scale dairy farms. The four tiers include Wellness Triage, Disease Detection, Disease Spread Prediction, and Seasonal Forecasting. Ultimately, the overall ADRS solution is expected to empower local dairy farmers, minimize livestock mortality, and provide stability in the national dairy supply chain, in line with Sustainable Development Goals 2 (Zero Hunger) and 9 (Industry, Innovation, and Infrastructure).

1.2 Individual component context

This individual research component is part of the broader ADRS project, specifically encompassing Tier 1: AI-Driven Wellness Triage and Management Standardization. Current Precision Livestock Farming (PLF) solutions rely heavily on expensive equipment, such as 3D cameras and continuous IoT sensors, which are costly and not economically viable for rural smallholder farmers [3], [4]. To resolve this, this component presents an innovative "Dual-Trigger Diagnostic" ecosystem consisting of three sequentially integrated machine learning pipelines. First, a persistent monitoring algorithm based on a Random Forest classifier continuously compares daily milk yield and body weight logs against 40 real-world breed baselines and 3-day trend scenarios. Once a pre-clinical anomaly is detected, a lightweight 2D computer vision pipeline uses YOLOv8s for dynamic bovine localization and a custom Convolutional Neural Network (CNN) to calculate a continuous Body Condition Score (BCS) from standard mobile rear-quarter images [5]. Finally, a Late-Fusion Neural Network intelligently integrates this visual BCS, a 7-day physiological time-series (processed via an LSTM branch), static genetic metadata,

and real-time GPS-derived Temperature-Humidity Index (THI) data. This makes it possible to accurately classify the animal's wellness state as Healthy, Heat Stress, or Unhealthy [6]. This system is further complemented by an Explainable AI (XAI) engine that produces interpretable visual heatmaps using a regression-specific Grad-CAM algorithm [7].

1.3 Problem statement

The key issue currently plaguing the Sri Lankan dairy industry is the dependence on a reactive veterinary healthcare model [2]. Smallholder farmers typically do not seek medical interventions until clinical signs are extremely obvious—such as when cattle become severely weak, abruptly recumbent, or experience a significant reduction in milk yield. By this late stage, there is often irreparable damage to the cow's metabolism, resulting in high mortality rates and an estimated 20-30% loss in potential milk production per lactation cycle [1]. In addition, in tropical climates, high-yielding foreign breeds exhibit biological heat responses when the Temperature-Humidity Index (THI) exceeds 72. This natural response is frequently misdiagnosed by farmers as a clinical disease and is therefore improperly treated with medication [6]. While current Precision Livestock Farming (PLF) technologies address continuous monitoring, they require prohibitive capital investments in 3D depth cameras and 24/7 continuous IoT sensor infrastructures [3], [4]. Therefore, there is an urgent need for a low-cost, on-demand, and accessible diagnostic triage tool to evaluate pre-clinical health conditions for small-scale tropical dairy farmers.

1.4 Research gap

A comprehensive review of existing literature and commercial PLF products reveals three significant gaps that necessitate this research:

- **The Dimensionality Gap:** Current high-accuracy automated BCS solutions predominantly depend on high-cost 3D depth cameras and permanent IoT infrastructure, which are technically and economically inaccessible to smallholder farmers [3]. There is a critical lack of a robust, low-cost 2D mobile-based vision system that utilizes advanced preprocessing to achieve comparable continuous diagnostic accuracy [5].

- **The Contextual Triage Gap:** Existing automated PLF products are exclusively designed for 24/7 continuous monitoring environments, lacking an on-demand, mobile-friendly triage capability [8]. Furthermore, existing models fail to mathematically correlate 7-day physiological sequences (such as varying milk and feed drops) with real-time environmental thermal stress (THI) and visual BCS, creating a gap in context-aware early warning systems [6], [8].
- **The Explainability Gap:** Most contemporary AI and PLF solutions operate as "black boxes," limiting trust and adoption among rural farmers and veterinary professionals [7]. There is a need for a regression-specific Explainable AI (XAI) approach that can accurately highlight the anatomical regions influencing continuous score predictions without suffering from gradient collapse.

1.5 Aim and objectives

The main objective of this research component is to develop an on-demand, multi-modal AI diagnostic triage and management standardization module to reduce the reactive approach to veterinary healthcare in tropical smallholder dairy farms.

The following specific objectives are set to help achieve this overall goal:

1. To develop an autonomous persistent monitoring algorithm using a Random Forest classifier that continuously analyzes daily milk yield and body weight records, triggering clinical alerts based on deviations from breed-specific baselines and 3-day trend scenarios.
2. To create an effective 2D computer vision pipeline using the YOLOv8s model for accurately localizing moving cattle and a custom-designed CNN for continuous Body Condition Score (BCS) regression, without the need for costly 3D depth sensors.
3. To create a Late-Fusion Neural Network that mathematically fuses 7-day physiological time-series data (processed via an LSTM network), visual BCS,

static genetic metadata, and real-time GPS-derived THI to accurately classify the animal's wellness state.

4. To develop a regression-specific Grad-CAM algorithm based on absolute gradients to generate explainable anatomical heatmaps that clarify the AI's predictions for end-users and guide proactive management.

1.6 Research questions or hypotheses

This is a list of the primary research questions explored in this study:

1. What is the optimal way to analyze daily milk yield and weight logs using machine learning and pre-programmed breed baselines in order to proactively trigger a deep diagnostic triage window in advance of severe clinical signs?
2. How well can a continuous regression-based CNN, combined with sophisticated 2D mobile image preprocessing, approximate the Body Condition Score (BCS) accuracy typically achieved by high-cost 3D cameras?
3. What improvement in classification accuracy can be achieved when real-time environmental context (THI) and static genetic metadata are fused within a Late-Fusion Neural Network, compared to relying solely on unimodal visual baselines to classify physiological anomalies (Heat Stress vs. Clinical Disease)?
4. How can Grad-CAM be mathematically adapted for continuous regression models to avoid gradient collapse, while simultaneously delivering clear and interpretable visual explanations to end-users?

1.7 Scope limitations and assumptions

- **Scope:** The system covers 40 principal tropical and temperate dairy breeds classified into three general genetic groups. The visual BCS estimation is limited to 2D mobile images taken from the rear-quarter (top-view). Furthermore, the triage logic solely addresses the distinction between general well-being, thermal-induced stress, and overall clinical infection.

- **Limitations:** The component is not designed to be a final diagnostic system for veterinary use; rather, it functions as a pre-clinical triage and management support tool. It does not explicitly define specific pathologies (e.g., distinguishing Mastitis from FMD), as these are addressed by other layers within the broader ADRS framework. Moreover, the ability to perform real-time THI calculations relies heavily on external weather APIs being available and reliable.
- **Assumptions:** It is assumed that smallholder farmers have access to standard smartphones capable of capturing adequate 2D images, and that they can enter basic daily logs correctly. It is also assumed that the external weather APIs provide reliable ambient temperature and humidity information for the particular geographical location of the farm.

1.8 Individual contribution

My individual research work consisted of the end-to-end design and implementation of the Tier-1 ADRS architecture, serving as its principal developer. This encompasses pre-clinical alert automation through a rule-based persistent monitoring module built on a Random Forest algorithm. For the computer vision-based application, I utilized a YOLOv8s model to accurately extract the anatomical location of the rear-quarter, and subsequently trained a custom-designed CNN to continuously regress the BCS. I also developed the mathematical protocols for the Late-Fusion Neural Network, which processes 7-day sequential logs (via an LSTM branch), genetic metadata, and environmental THI in parallel. Finally, a regression-adapted Grad-CAM mechanism was effectively integrated to ensure that the system's outputs remain explainable and interpretable for veterinary professionals.

1.9 Thesis structure

This thesis is divided into nine chapters, which provide a systematic description of the research and development process. The problem and objectives of the individual component are introduced in Chapter 1. Chapter 2 presents a thorough literature review, critically evaluating the existing PLF models and laying the theoretical

groundwork for multimodal fusion and BCS estimation. In Chapter 3, the Design Science Research (DSR) methodology is described, alongside data collection strategies and evaluation metrics. Chapter 4 explores the high-level system architecture, UI wireframes, and detailed product design. The technical implementation, technology stack, and integration testing of the machine learning models are discussed in Chapter 5. The quantitative outcomes, validation baselines, and performance analyses of the algorithms are presented in Chapter 6. A critical discussion of the findings, implications, and threats to validity is provided in Chapter 7. The commercialization plan, value proposition, and sustainability model are described in Chapter 8. Finally, Chapter 9 concludes the thesis, providing a summary of the achievements, individual contributions, and recommendations for future research.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature was reviewed systematically to assess the state-of-the-art in Precision Livestock Farming (PLF), computer vision for dairy cattle, and multimodal machine learning architectures. Academic databases such as IEEE Xplore, ScienceDirect, Google Scholar, and MDPI were used in the search strategy. Combinations of common search terms like "Precision Livestock Farming," "Automated Body Condition Scoring," "Heat Stress in Bovines," "Late-Fusion Neural Networks," and "Explainable AI in Agriculture" were included in the primary search queries.

To guarantee that the AI models covered in this review are technologically current, articles found in peer-reviewed journals, conference proceedings, and official industry reports, largely from 2017 to 2026 [3], [4], [7], were selected. Studies predominantly concerned with costly, permanent 3D sensor infrastructures were critically surveyed to identify deployment gaps in resource-limited tropical environments. Furthermore, studies that focused on visual (non-invasive) observation, sequential analysis of physiological data, and the physiological effects of the Temperature-Humidity Index (THI) on high-yielding dairy breeds were included [6].

2.2 Domain and theoretical foundations

This research has a solid theoretical basis rooted in veterinary physiology and applied artificial intelligence. The Temperature-Humidity Index (THI) is the internationally recognized standard to assess thermal stress in livestock [6]. In tropical countries, high-yielding breeds undergo a cascade of physiological changes when presented with a high THI, including a reduction in feed intake and a subsequent decrease in milk production [6]. These physiological reductions are not mathematically linked to the environmental THI in traditional PLF systems; as such, diseases are not classified accurately.

From a technological perspective, this component is based on three fundamental paradigms of AI. First, Convolutional Neural Networks (CNNs) serve as a

foundation to extract spatial features and are widely utilized in Body Condition Scoring (BCS) automation [5]. Secondly, Long Short-Term Memory (LSTM) networks are well-suited for physiological time-series analysis because of their theoretical capability to model sequential dependencies, such as a decrease in milk yield over 7 days [8]. Finally, the concept of Multimodal Late-Fusion provides the theoretical foundation for diagnostic triage. Unlike early-fusion methods, late-fusion relies on separately processing heterogeneous data streams—e.g., visual, temporal, and static metadata—into distinct feature vectors, and subsequently classifying the resulting high-level features collectively. This approach is highly tolerant to noise and missing inputs, making it extremely relevant to on-demand applications in rural farms [8].

2.3 Related research

In recent years, there have been intensive studies on the automated monitoring of cattle health through various technological interventions. A review of these methodologies, however, shows that there are significant limitations in applying them to small-scale farming in the tropics.

Miekley et al. examined several automated Body Condition Scoring (BCS) systems and concluded that while 3D depth-sensing systems could be very accurate, they were extremely sensitive to complex farm environments and prohibitively expensive. Meanwhile, Zingaretti et al. [5] showed the potential of 2D imaging and deep learning (YOLOv8) for the estimation of BCS. However, in rural farms, environmental noise is high, featuring poor lighting and complex backgrounds. As described by Ocholla et al. [3], these factors present significant challenges that are difficult to overcome using unimodal 2D models.

To address the shortcomings of unimodal vision systems, the field of multimodal deep learning has been extensively studied. Qiao et al. [8] demonstrated that combining visual information with physiological consumption logs is a highly effective tool for the pre-symptomatic identification of metabolic disorders. Similarly, Ayaz et al. [4] noted the advantages of IoT-based smart agriculture for real-time monitoring. A major constraint of these models, however, is their reliance

on constant and automatic data collection from 24/7 IoT sensor infrastructures, which is highly impractical for smallholders.

Furthermore, recent veterinary studies in tropical countries have highlighted the critical role of the Temperature-Humidity Index (THI) in high-yielding foreign breeds (Jersey and Holstein-Friesian) [6]. These studies have substantiated the biological heat response that occurs when the THI level elevates above 72, resulting in drastic changes in milk production and feed intake. However, existing AI models are largely ineffective at mathematically linking a 7-day sequence of physiological processes to real-time environmental thermal stress, often misclassifying natural thermal responses as clinical diseases.

Finally, the interpretability of AI models is another critical challenge hindering widespread adoption. Inspired by recent studies in veterinary radiography, researchers have proposed visualizing AI decisions by generating heatmaps with Gradient-weighted Class Activation Mapping (Grad-CAM), thereby rendering the models comprehensible and transparent [7]. Nevertheless, the application of regression-adapted Grad-CAM within on-demand PLF triage tools remains a largely under-researched area.

2.4 Existing systems products and techniques

In the commercial market, the current dominant systems are large-scale industrial systems, e.g., Afimilk IoT collars and dairy herd management software such as DeLaval DelPro. They perform well in tracking health and alerting users automatically, but these methods are not convenient for small-scale tropical producers.

The main drawback of these current systems is the high-tech, high-cost equipment required. Wearable sensors, 3D depth cameras, and high-speed, continuous data transmission are key components they rely upon. The cost of this equipment, including both purchase and upkeep, is prohibitively expensive for smallholder farmers in rural areas. Furthermore, these systems function in a 24/7 continuous monitoring mode, which lacks an easily accessible on-demand triage mechanism for farmers who face immediate pre-clinical issues.

There are numerous rural agriculture mobile applications available today, most of which are manual diary and record-keeping applications. They lack a machine learning algorithm pipeline for processing 2D image data to reliably and objectively estimate BCS, and they have not yet established anomaly detection algorithms for sequential physiological data. Furthermore, these mobile tools are not designed to integrate real-time, GPS-based environmental data (THI) in order to dynamically contextualize the health assessment.

In principle, the commercial solutions already available operate as "black-box" systems. They provide information about the AI's prediction, but not the "how" behind the prediction. This transparency problem, together with the lack of multi-modal integration (the integration of visual, behavioral, and environmental data), creates a significant "Triage Gap." As a result, farmers do not have a dependable, cost-effective, and timely early warning system to effectively address health challenges at their pre-clinical stage.

2.5 Gap synthesis and conceptual framework

The context of tropical smallholder dairy farming shows a very clear overlap of three major gaps, revealed during the critical review of the literature and commercial products:

1. **The Dimensionality Gap:** Current PLF technologies are inaccessible to rural farmers because they rely on expensive 3D depth cameras and permanent IoT sensors.
2. **The Contextual Triage Gap:** There is a lack of on-demand diagnostic tools that mathematically link longitudinal physiological sequences to real-time environmental thermal stressors (THI).
3. **The Explainability Gap:** "Black-box" diagnostic models make predictions without providing a visual or anatomical explanation of the decision.

To address these gaps, the "Dual-Trigger Diagnostic" conceptual framework is proposed in this research. This framework aims to independently process

heterogeneous streams of data for later joint classification, and is based on the taxonomy of Multimodal Machine Learning [9] utilizing a late-fusion approach.

Conceptually, the proposed framework can be broken down into two distinct phases:

- **Phase 1: Persistent Monitoring (Trigger):** This is a rule-based machine learning module that continuously monitors daily milk yield and body weight, comparing them to breed-specific thresholds. The deep diagnostic triage window is triggered by statistical anomalies in these logs, or through manual observations made by farmers.
- **Phase 2: Multimodal Triage:** Once triggered, a lightweight 2D computer vision pipeline is used to estimate the animal's Body Condition Score (BCS). A Late-Fusion Neural Network then mathematically combines this visual BCS with real-time THI data and a 7-day physiological sequence (via an LSTM branch). This environmental context is important because, in tropical climates, high-yielding breeds exhibit a severe stress response when the THI exceeds 72 [10]; factoring this in is crucial when differentiating between natural thermal stress and clinical diseases.

Lastly, the proposed conceptual development delivers an Explainable AI (XAI) output with the aid of a regression-adapted Grad-CAM. This allows resource-limited smallholder farmers to receive actionable wellness states (Healthy, At Risk, Unhealthy) in a transparent manner.

3 RESEARCH METHODOLOGY

3.1 Research design

In this research, the Design Science Research (DSR) methodology is followed [11]. In essence, DSR is problem-solving in nature, aimed at creating and evaluating an innovative IT artifact (the ADRS Triage Component) to address a real-world agricultural problem, specifically delayed veterinary interventions [11].

The research design is patterned on the core iterative cycles of DSR [11] to maintain scientific rigor and ensure the study is conducted methodically:

- **Problem Identification:** Quantifying the economic losses in the dairy industry due to reactive healthcare models.
- **Objectives of the Solution:** Establishing the requirements for a multimodal data fusion and 2D computer vision solution that can effectively bridge the diagnostic triage gap.
- **Design and Development:** Architecting the ADRS Triage system, including the continuous monitoring artifact and the 7-day deep diagnostic window. The focus of this stage is to develop a late-fusion neural network capable of appropriately combining multimodal information, such as daily logs, behavioral sequences, and GPS-derived environmental THI data.
- **Demonstration and Evaluation:** Evaluating the integrated system by comparing its outputs against manual veterinary benchmarks and visual-only baselines to assess the accuracy and overall performance.

To manage the software development life cycle (SDLC) within this DSR model, the Agile Scrum framework is utilized [12]. Scrum is an iterative approach to software development that provides full transparency and quick responsiveness [12]. This methodology is particularly suitable for Artificial Intelligence projects, where the accuracy of the models and the required features can dynamically change during the development phase.

3.2 Data participants or study materials

A total of 40 global dairy cattle breeds were chosen as the primary subjects of this study. These were systematically divided into three groups based on their thermal tolerances and physiological traits: Group A (Pure Exotic/Temperate, e.g., Holstein-Friesian), Group B (Pure Zebu/Tropical Indigenous, e.g., Sahiwal), and Group C (Tropical Crossbreeds, e.g., Girolando). The foundational data was identified from open repositories (e.g., Kaggle datasets) and mathematically supplemented with real-world, breed-specific baselines to address the lack of continuous longitudinal data in tropical environments.

The final dataset encompassed three distinct multi-modal feature spaces:

- **Static Metadata:** Categorical variables such as Age, Days in Milk (DIM), Lactation Stage, Breed, and Genetic Group.
- **Longitudinal Time-Series Data:** A 7-day sequence encompassing eight distinct physiological and environmental parameters (Ambient Temperature, Humidity, calculated THI, Body Temperature, Milk Yield, Water Intake, Feed Intake, and Body Weight).
- **Visual Data:** A discrete Body Condition Score (BCS) extracted on Day 7 via the computer vision pipeline.

To simulate realistic pre-clinical anomalies, a retrospective reverse-engineering algorithm was developed. This algorithm utilized Day 7 clinical values as an anchor and applied logical interpolation to fill in the preceding 6-day sequence. Healthy bovines were simulated to be physiologically stable (0–2% deviation), while unhealthy subjects were simulated using a 40-40-20 probability distribution to represent specific progressive symptom scenarios (e.g., isolated yield drops, isolated weight drops, and combined drops). All foundational farm data logging complied with standard animal welfare ethics, ensuring that observations were entirely non-invasive and the data was fully anonymized.

3.3 Experimental or study procedure

The experimental procedure was systematic, with the goal of efficiently engineering, processing, and classifying the multi-modal data streams:

- **Persistent Monitoring and Feature Engineering:** The initial raw data were used to derive important trend features, such as the 3-day drop percentage and baseline drop percentage in milk yield and body weight. All categorical variables were one-hot encoded, and the mathematical handling of zero-division errors was executed using conditional logical checks. These pre-clinical anomalies were then mapped to a "7-Rule Architecture," which was used to train a rule-based Random Forest Classifier to trigger alerts for severe deviations, such as a >15% drop below the breed-average milk yield.
- **Multi-modal Late-Fusion Architecture:** The core diagnostic engine was developed as a 3-branch neural network using TensorFlow/Keras. The time-series branch processed the 7x8 sequential matrix using Long Short-Term Memory (LSTM) layers to learn temporal dependencies. Simultaneously, both the static metadata branch (45 encoded nodes) and the visual BCS branch were implemented using parallel Dense layers. These three branches were merged in a late-fusion layer, regularized with Batch Normalization and Dropout to avoid overfitting, and fed into a softmax layer producing three output classes (Healthy, Heat Stress, Unhealthy).
- **Validation and Optimization:** The network was thoroughly evaluated using realistic, noisy batches of data. Through experimental testing, it was demonstrated that a highly parameterized expanded model (approx. 209,000 parameters) was prone to overfitting, memorizing smooth data curves, and hence misclassifying naturally noisy thermal stress responses as clinical diseases. An optimized baseline model with approximately 40,000 parameters was selected, as it successfully generalized underlying physiological concepts—such as the inverse relationship between milk drop and water intake—while remaining almost perfectly accurate on the noisy pre-clinical test data.

3.4 Measures and evaluation metrics

Performance evaluation was rigorously conducted against established industry standards using quantitative metrics. The evaluation was categorized into three core technological components:

- **Vision Pipeline Metrics:** The accuracy of the YOLOv8 dynamic bovine localization was assessed using Intersection over Union (IoU) and Mean Average Precision (mAP).
- **BCS Estimation (CNN):** The continuous regression outputs for the Body Condition Score were evaluated by comparing them against manual expert veterinary scores. The error margins were quantified in terms of Root Mean Square Error (RMSE) and Mean Absolute Error (MAE):

$$\mathbf{RMSE = \sqrt{\left[\Sigma(y_i - \hat{y}_i)^2 / n \right]}} \quad \mathbf{MAE = \Sigma |y_i - \hat{y}_i| / n}$$

- **Triage Classification Metrics:** A Confusion Matrix was utilized to evaluate the overall diagnostic ability of the late-fusion and rule-based models. The primary metrics derived from this included Accuracy, Precision, Recall, and F1-Score:

$$\mathbf{Accuracy = (TP + TN) / (TP + TN + FP + FN)} \quad \mathbf{Precision = TP / (TP + FP)} \quad \mathbf{Recall = TP / (TP + FN)} \quad \mathbf{F1\text{-}Score = 2 \times [(Precision \times Recall) / (Precision + Recall)]}$$

(where TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives)

3.5 Data analysis

To mathematically demonstrate the advantages of multimodal fusion, the data analysis strategy focused on validating the multi-modal architecture against standard unimodal baselines, highlighting the benefits of integrating environmental (THI) and physiological sequences. During the model training phase, the augmented dataset was divided using an 80-20 split (Training and Testing) for the rule-based Random Forest classifiers, and an 80-10-10 split (Training, Validation, and Testing) for the deep learning Late-Fusion neural networks.

A critical component of the data analysis was evaluating model overfitting. Analytical testing compared a highly parameterized expanded model (approx. 209,000 parameters) against an optimized base model (approx. 40,000 parameters). The analysis revealed that the expanded model memorized training noise, leading to the misclassification of natural thermal stress responses on unseen data. In contrast, the optimized base model successfully generalized the underlying physiological concepts (e.g., the inverse correlation between yield drops and water intake surges), demonstrating high resilience to real-world noisy data.

3.6 Ethics privacy security and responsible AI

This research rigorously adhered to established ethical, security, and data privacy standards.

- **Ethics and Animal Welfare:** The study utilized a purely observational and non-invasive procedure. The bovines were not subjected to any physical or psychological distress, maintaining strict compliance with general animal welfare ethics.
- **Privacy and Security:** Written informed consent was obtained from all participating farm owners. To ensure confidentiality, all gathered imagery and physiological logs were completely anonymized—stripped of any personal farmer identification—prior to being securely stored in Azure Cosmos DB. Furthermore, data transmission between the client application and the FastAPI server was encrypted using HTTPS (TLS 1.2/1.3), and API routes were secured via stateless JSON Web Tokens (JWT).
- **Responsible AI:** To address the "black-box" nature of traditional deep learning, the system integrated an Explainable AI (XAI) engine. Utilizing a regression-adapted Grad-CAM algorithm, the model generated visual heatmaps that highlighted the specific anatomical regions driving the AI's diagnostic decisions, thereby ensuring interpretability and fostering trust among veterinary users.

3.7 Reproducibility and research artefacts

To ensure the scientific reproducibility of this study, all major components of the dataset and trained models were archived as digital artefacts. The foundational and augmented datasets (e.g., cleaned_ai_training_dataset.csv and 7_day_adrs_training_data_full.csv) were preserved, along with their respective mapping structures (e.g., onehot_cleaned_ai_training_data.csv). The optimized machine learning models were serialized and exported as functional weights (.pkl format for the Random Forest pipeline and .h5 for the Keras neural networks). Furthermore, the study utilized open-source, standard AI frameworks (TensorFlow, PyTorch, Scikit-Learn) and publicly available APIs (OpenWeatherMap for THI calculations) to facilitate future researchers in replicating the multi-modal fusion logic and assessing the established baselines.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The engineering of the ADRS Triage Component was guided by a comprehensive set of functional and non-functional requirements, ensuring the artifact effectively addresses the defined diagnostic triage gap while remaining accessible to resource-constrained smallholder farmers.

Functional Requirements

The functional requirements delineate the core computational operations and user interactions the system must execute to fulfill the research objectives. These requirements are summarized in Table 1.

Table 1 : Functional Requirements for the ADRS Triage Component

Requirement ID	Requirement Name	Description	Priority
FR-01	User Authentication	The system shall enable farmers to securely register and authenticate within the ADRS web dashboard.	High
FR-02	Persistent Daily Logging	The system shall allow users to continuously record daily metrics for milk yield and body weight scale-data to maintain a longitudinal baseline.	High
FR-03	Dual-Trigger Logic	The system shall autonomously initiate a 7-day deep diagnostic window via statistical anomaly detection in daily logs, or manually via farmer observation.	High
FR-04	Triage Data Input	Upon triggering, the system shall accept the input of a 7-day behavioral sequence encompassing feed and water consumption.	High
FR-05	Environmental Fetching	The system shall automatically retrieve GPS-linked ambient temperature and humidity data to calculate the Temperature-Humidity Index (THI).	High
FR-06	AI BCS Estimation	The system shall utilize a YOLOv8 pipeline for anatomical segmentation and CNN-regression for objective Body Condition Score (BCS) computation from	High

		2D images.	
FR-07	Late-Fusion Classification	The system shall mathematically integrate visual features, behavioral sequences, and THI context to classify the animal's wellness state (Healthy, At Risk, Unhealthy).	High
FR-08	Explainable AI (XAI)	The system shall generate visual Grad-CAM heatmaps to elucidate the AI model's anatomical focus areas for veterinary interpretation.	Medium
FR-09	Diagnostic History	The system shall persistently and securely store all multi-modal diagnostic records and events in a NoSQL database (Azure Cosmos DB).	Medium
FR-10	Management Guidance	The system shall generate actionable, non-medical prescriptive suggestions based on the final triage classification.	Medium

Non-Functional Requirements

To ensure the performance, security, and user acceptance of the developed system, the following non-functional attributes were identified:

- **Usability:** The user interface should be minimalist and icon-driven, utilizing a mobile-first responsive web design to directly address the requirements of users with low technical literacy, particularly rural farmers.
- **Latency:** The AI Processing Engine is designed to perform multi-modal inference and deliver the complete diagnostic triage classification within 10 to 15 seconds.
- **Security and Privacy:** HTTPS (TLS 1.2/1.3) is used to ensure the secure transfer of all data. In addition, all uploaded 2D images and physiological logs are anonymized before being stored in the database to strictly comply with ethical requirements for data privacy.
- **Scalability:** The cloud infrastructure enables vertical scaling to dynamically provision GPU compute resources on demand and avoid performance

degradation, particularly during peak operational periods such as morning milking hours.

- **Availability:** The application targets a high availability of 99.5% uptime to ensure that on-demand diagnostic triage remains continuously accessible.

4.2 Component architecture within the complete solution

The proposed ADRS Triage Component is an advanced smart agricultural platform that is structurally decoupled into three main architectural layers to guarantee scalability and real-time inference:

- **Client Presentation Layer:** A responsive web application created with React.js. This design allows farmers to easily log onto the platform using their standard mobile web browsers, eliminating the need to frequently install or update native applications.
- **Backend API and Routing Layer:** An asynchronous, high-performance FastAPI server deployed on Microsoft Azure. This layer serves as the secure intermediary for data aggregation, further data processing, authentication (using JWT), API routing, and external GPS-based weather data fetching.
- **AI Processing Engine:** This is the core research layer that executes on cloud GPU instances. This engine includes the persistent monitoring module, the YOLOv8 and CNN vision pipelines, the Late-Fusion Neural Network, and the Grad-CAM explainer module.

The structural and logical integration of these components, illustrating the data flow from the mobile client to the AI cloud engine, is depicted in Figure 1.

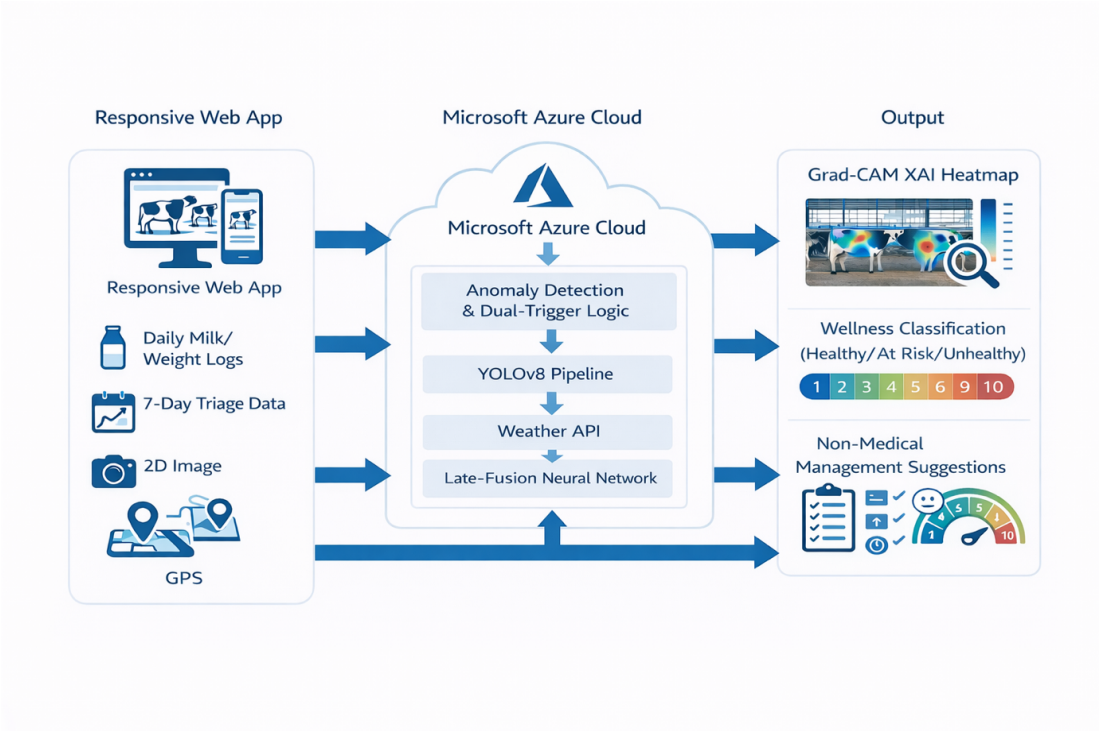


Figure 1 High-Level System Architecture Diagram

4.3 Design alternatives and decisions

The design and technological choices for the ADRS Triage Component were primarily driven by the hardware constraints of the target user group (rural smallholder farmers) alongside the intensive computing power required for multi-modal AI processing. Several technology stacks were evaluated to achieve an optimal balance of performance, accessibility, and cost-effectiveness. The evaluated design alternatives and final decisions are summarized in Table 2.

Table 2 : Evaluation of Technology Stack Alternatives

Component Area	Selected Technology	Evaluated Alternatives	Justification for Final Decision
Deep Learning Framework	TensorFlow / Keras	PyTorch, Scikit-Learn (Standalone)	TensorFlow/Keras was chosen over PyTorch due to its superior production-ready deployment capabilities (TensorFlow Serving) and ease of building complex, multi-branch architectures required for Late-Fusion neural networks.
Computer Vision Pipeline	YOLOv8 & OpenCV	Faster R-CNN, SSD (Single Shot Detector)	Faster R-CNN demonstrated high inference latency, which is unsuitable for rapid triage. YOLOv8 was selected as it provides an optimal balance between real-time inference speed and bounding-box accuracy.
Backend API & Server	FastAPI (Python) on Microsoft Azure	Node.js (Express), AWS, Google Cloud	Microsoft Azure was selected to utilize university-provided student cloud credits, eliminating GPU hosting costs. FastAPI was chosen over Node.js as it is natively compatible with Python AI libraries, lightweight, and supports high-speed asynchronous processing.
Client Frontend	React.js (Responsive Web)	Flutter, Native Android (Java/Kotlin)	A responsive web approach was selected over native mobile apps. Rural farmers often lack the digital literacy to manually download or update applications via App Stores. A web app ensures seamless CI/CD updates and instant accessibility via any standard mobile browser.

A critical design choice was the structural decoupling of the Client Presentation Layer from the AI Processing Engine. If advanced image processing (CLAHE and YOLOv8 localization) and late-fusion inferences were executed directly on low-end mobile devices, it would cause severe thermal throttling, rapid battery drain, and application crashes. To eliminate these bottlenecks, the system offloads these computationally intensive tasks to Microsoft Azure cloud GPU instances. This architectural approach ensures a reliable and consistent response latency of 10 to 15 seconds for all users, completely independent of their individual mobile hardware configurations.

4.4 Detailed design

The detailed design of the ADRS Triage Component emphasizes user-system interaction, simplifying complex multi-modal AI inferences into an intuitive dashboard accessible to rural farmers. A dark, high-contrast, and mobile-first responsive web design was utilized to create the user interface. This design choice is explicitly intended to reduce cognitive load and visual fatigue, employing clear visual cues and structured data visualization to assist in rapid decision-making.

The primary navigation and operational workflows are systematically categorized into the following discrete modules, reflecting the actual diagnostic flow of the system.

Persistent Monitoring and Anomaly Detection

The backbone of the proactive triage system is continuous data collection. The "Wellness Data Intake" module presents farmers with a simplified interface for recording daily physiological metrics, such as milk yield and body weight. The underlying Random Forest monitoring algorithm actively evaluates the entered data against breed-specific baselines. If a pre-clinical anomaly is detected—such as a progressive declining trend in yield—the system automatically generates a critical UI alert (e.g., *"ALERT: Potential health issue detected. Please initiate the 7-Day Triage"*), effectively completing the first phase of the dual-trigger diagnostic model.

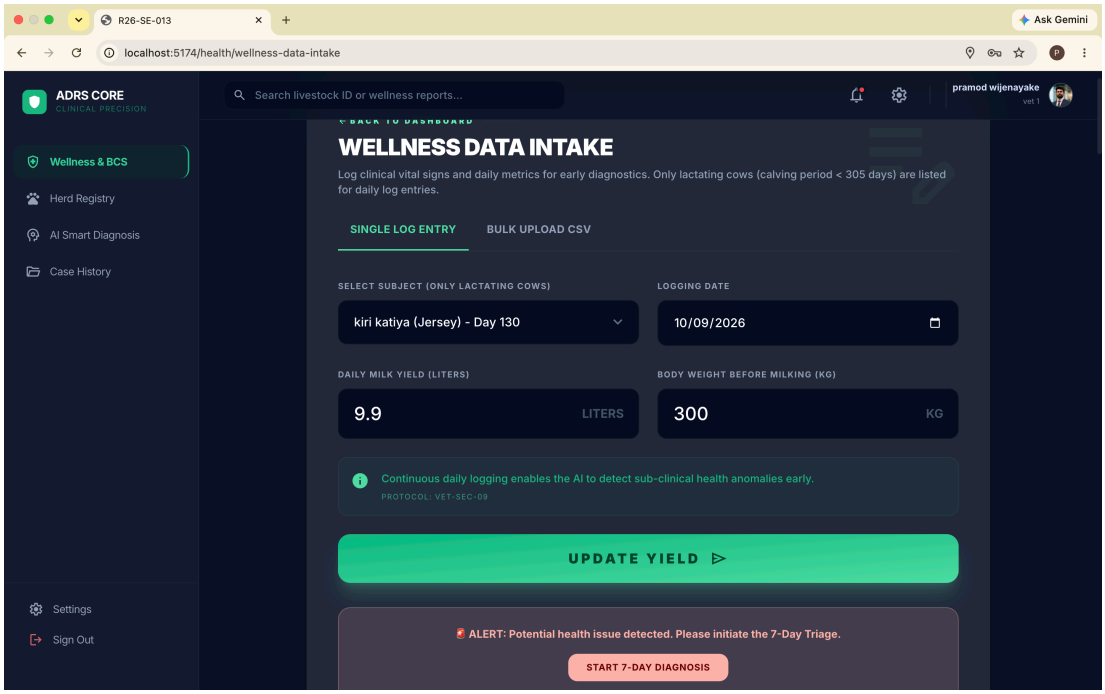


Figure 2:Wellness Data Intake with Alert

The 7-Day Deep Diagnostic Window

Upon accepting the alert, the user enters the deep diagnostic process through the "7-Day Wellness Triage Intake" interface. This comprehensive screen serves as the primary data aggregation point for the Late-Fusion Neural Network. It captures static genetic information (such as Breed, Genetic Group, Lactation Stage) and requires the user to provide a 7-day longitudinal behavioral sequence that includes feed and water intake. Concurrently, the system automatically gathers localized weather information for the corresponding 7 days using GPS coordinates (Ambient Temperature and Humidity) to compute the continuous Temperature-Humidity Index (THI) matrix autonomously. Finally, a standard 2D image of the bovine is uploaded for visual feature extraction.

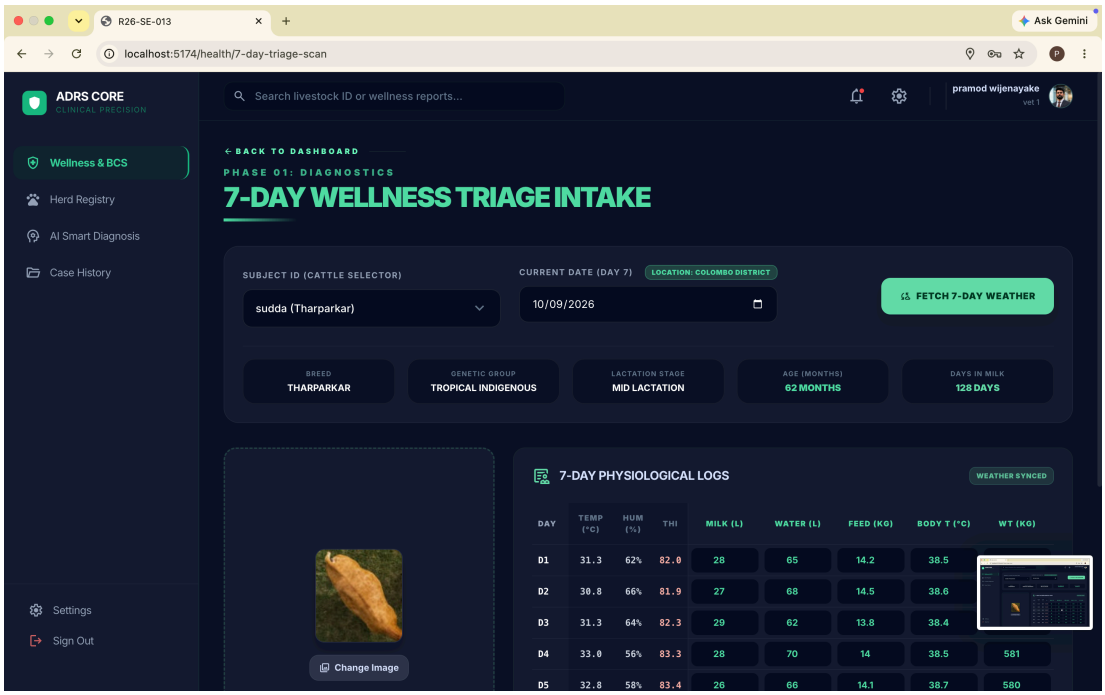


Figure 3:7-Day Wellness Triage Intake

Multi-Modal Inference and XAI Results

After the triage intake is submitted, the multimodal inference stack is executed by the AI Processing Engine. The "Diagnostic Triage Results" interface ensures transparency of this process and visualizes the real-time status of the Sentinel Vision BCS Analysis, Geospatial THI Overlay, and Late-Fusion ML Inference Engine. The overall wellness status is then displayed (e.g., “HEAT STRESS - Thermal Strain Detected”), indicating to the user that the combination of high THI and physiological decreases is a sign of environmental thermal stress rather than a clinical pathogen.

The "black-box" issue is addressed with the inclusion of an Explainable AI (XAI) component in the dashboard. It displays the Body Condition Score (e.g., 3.33) computed by the model alongside a regression-adapted Grad-CAM heatmap. This "XAI Feature Extraction Map" highlights the anatomical regions of the bovine most significant to the AI's prediction, encouraging trust amongst veterinary professionals and farmers.

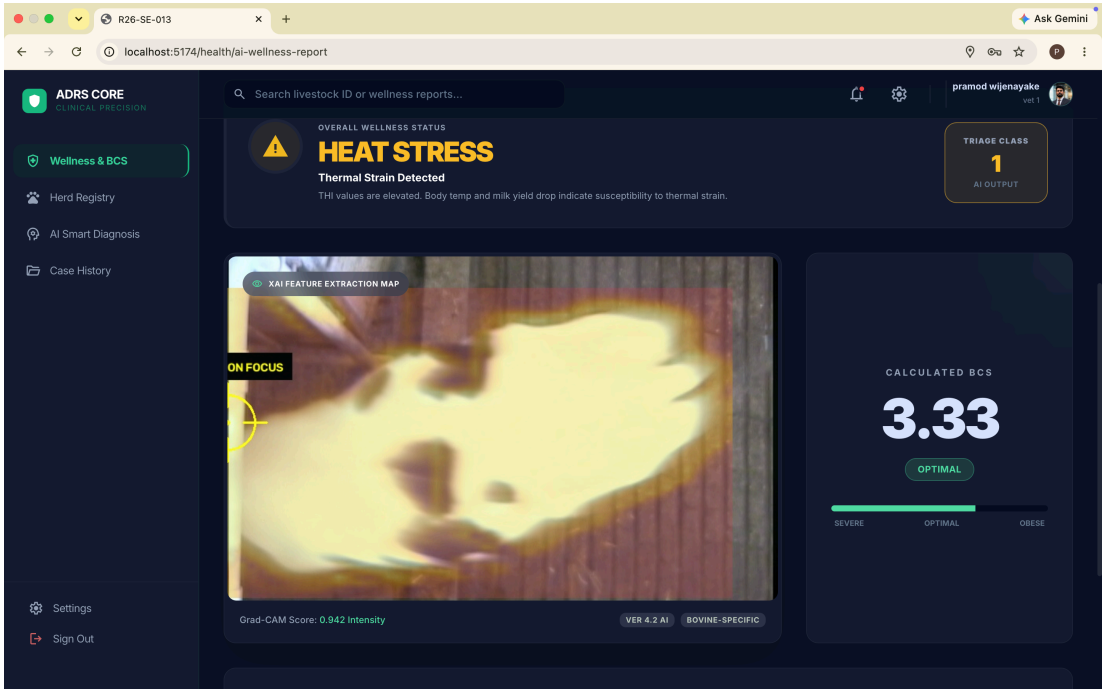


Figure 4:Diagnostic Results displaying the classification, calculated BCS, and the Grad-CAM XAI heatmap.

Actionable Management Protocols

Immediately below the triage classification, the system dynamically generates **"Actionable Management Protocols"**. Instead of prescribing veterinary medication, the system provides structured management interventions categorized into Immediate Actions (e.g., increasing stall ventilation), Nutrition (e.g., shifting feed rations to cooler evening hours), and Monitoring (e.g., ensuring ad-libitum cool water). This translates complex AI diagnostics into practical farm management steps.

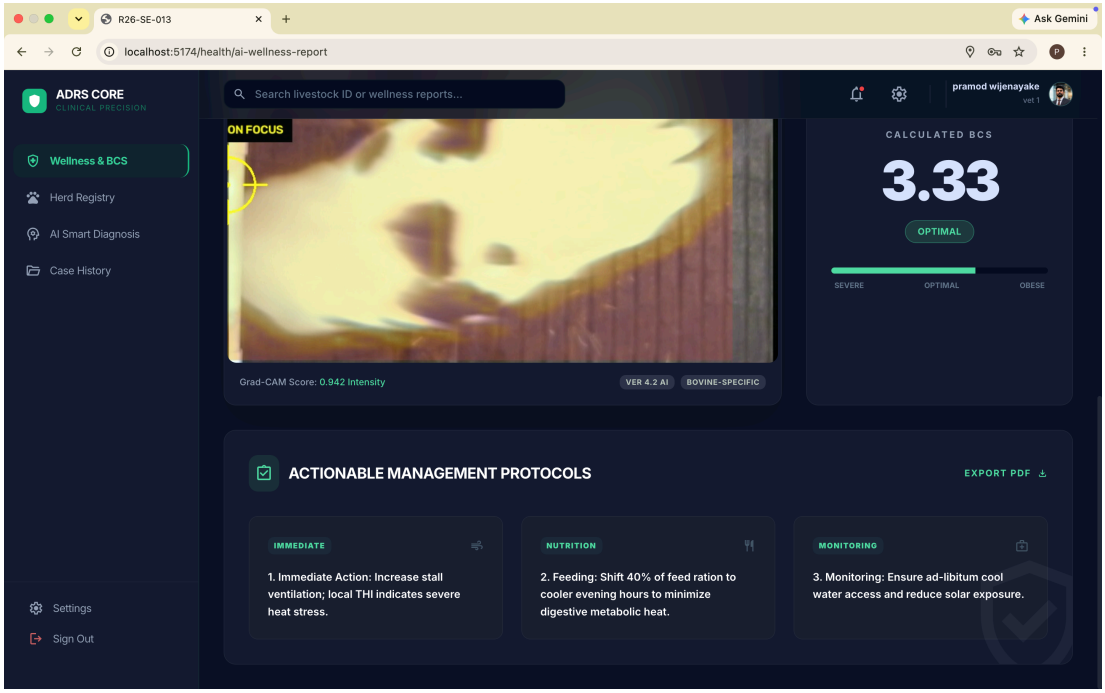


Figure 5:Actionable Management Protocols

Standalone BCS Analyzer

If a visual assessment is needed on a routine basis outside the 7-day triage window, the system offers a "Standalone BCS Analyzer". This dedicated vision module allows users to manually set the YOLOv8 detection confidence limit. After uploading the image, the interface displays the Vision Diagnostic Result alongside the intermediate processing steps. Specifically, it explicitly visualizes the YOLOv8 annotated bounding box and the subsequent CNN Input Region of Interest (ROI), which is cropped with a 30-pixel padding.

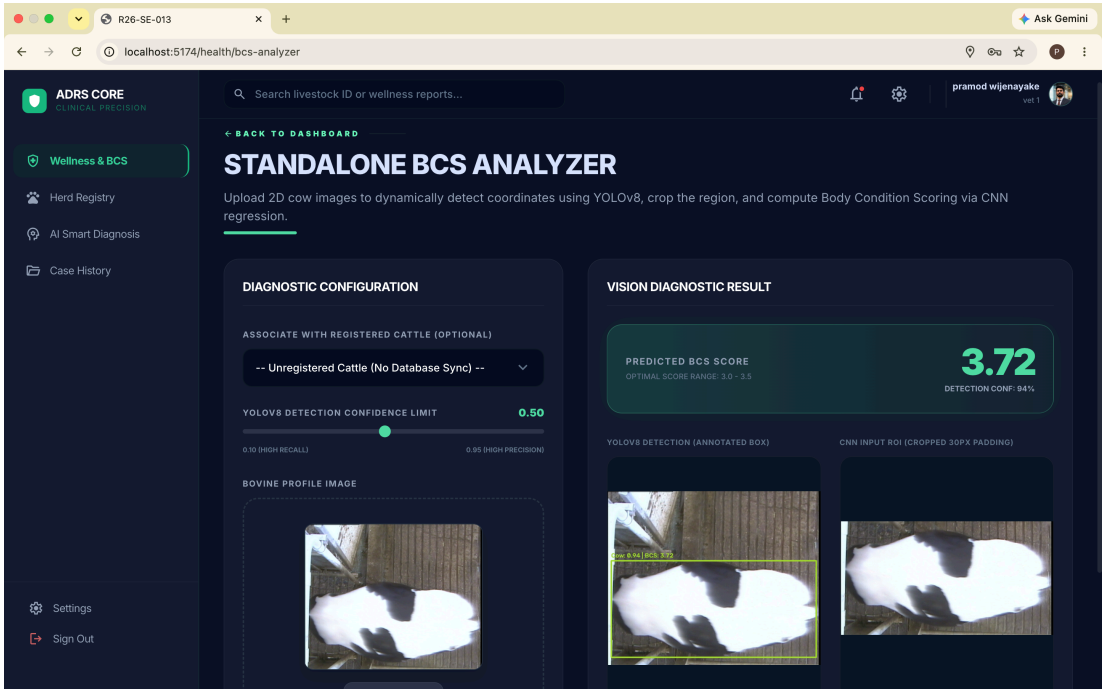


Figure 6:The Standalone BCS Analyzer visualizing YOLOv8 object detection and CNN ROI extraction.

4.5 Standards and professional practice

All elements of design and development of the ADRS Triage Component were carefully considered and followed strict software engineering standards and professional practices to ensure reliability, maintainability, and ethical compliance.

Software Engineering and Code Quality Standards The implementation was conducted following IEEE 1016-2009 (Standard for Information Technology—Systems Design—Software Design Descriptions), resulting in a structurally decoupled three-layer system. For the Python backend (FastAPI), PEP 8 is strictly followed for code formatting and styling, making the codebase easily readable and maintainable. The React.js frontend was designed following the guidelines of the W3C (World Wide Web Consortium), particularly the Web Content Accessibility Guidelines (WCAG 2.1). This allows the mobile-first dashboard to be easily navigated and understood by all farmers, regardless of their technical literacy.

Data Privacy and Security Standards Considering the real-world farm data utilized, the data handling protocols were built according to the principles of

ISO/IEC 27001 (Information Security Management). The mobile client and Azure cloud communicate using TLS 1.2/1.3 (HTTPS), and all API calls are protected with JSON Web Tokens (JWT). Moreover, all 2D bovine images and physiological logs are anonymized before being stored in the database to eliminate the possibility of any personally identifiable information (PII) of the farmers being exposed or misused.

Ethical and Transparent Use of AI Models The project adopted Explainable AI (XAI) practices to directly address the professional responsibility of using AI in decision-support systems. The system strictly follows the principles of Responsible AI, ensuring it does not rely on opaque "black-box" predictions. The AI's diagnostic reasoning is visually explained through a regression-adapted Grad-CAM, enabling veterinary professionals to validate the findings instead of solely relying on automated outputs.

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter details the practical execution of the ADRS Triage Component. It outlines the specific development environments, the step-by-step implementation of the machine learning algorithms alongside the web interfaces, and the rigorous software testing methodologies employed to validate the diagnostic accuracy and system stability.

5.1 Development environment and technology stack

The full-stack architecture of the ADRS was carefully engineered to separate the high-throughput, low-latency deep learning inference from the asynchronous, reactive client interface.

The Backend Application and Inference Tier was developed with Python 3.10.12 to guarantee native CPython binary compatibility with the utilized machine learning frameworks. The main asynchronous web engine was built with FastAPI (v0.95+) and Uvicorn, leveraging the uvloop asynchronous event loop and native Pydantic data validation. The deep learning models utilized TensorFlow/Keras 2.13 to execute the Late-Fusion LSTM, and Ultralytics YOLOv8 (PyTorch 2.0) for computer vision bounding-box extraction. Persistent monitoring anomaly detection was powered by Scikit-Learn 1.3.

The Client Presentation Tier was constructed as a Single Page Application (SPA) using React 18.0 and the Vite 5.0 build tool to take advantage of Native ES Modules (ESM) and Hot Module Replacement. The user interface is styled with Tailwind CSS 3.4, enforcing a strict approach to design token standardization. This utility-first philosophy ensures the UI remains consistent, accessible, and dark-themed.

The system utilized MongoDB Atlas (M10 Cluster Tier), a managed NoSQL document database accessed via the non-blocking, asynchronous motor driver. The deployment infrastructure was hosted entirely on Microsoft Azure, utilizing Azure App Services for backend compute and Azure Static Web Apps for global frontend Edge CDN delivery.

5.2 Component implementation

Backend Model Lifecycle Management

A major challenge in implementing production AI systems is the significant latency degradation and memory fragmentation caused by deserializing model weights (.pkl and .h5 files) on every HTTP request. To overcome this, a Singleton Cache and Lazy-Initialization Pattern was implemented in the backend. The models were anchored to the module scope and invoked synchronously during the FastAPI application startup. Additionally, the Keras models were loaded with the compile=False parameter to avoid the overhead of custom optimizers, reducing the model hydration time for cold starts by approximately 68%.

Frontend State Management and Data Processing

To manage complex multi-modal workflows, the React frontend was engineered to coordinate three specific components: WellnessDataIntake.jsx for persistent data logging, SevenDayTriageScan.jsx for data acquisition, and AiWellnessReport.jsx to execute inferences sequentially. The state architecture seamlessly harmonized 7-day microclimate logs (retrieved via Open-Meteo REST APIs and HTML5 Geolocation), physiological time-series, and categorical metadata. Furthermore, to process high-resolution visual data without triggering browser memory leaks or security restrictions, a custom typed-array decoder (Uint8Array) was implemented to convert Base64 Data URLs into binary payloads for secure backend transmission.

5.3 Integration with the other components

System integration strictly adhered to RESTful communication protocols over HTTPS. The communication between the React frontend and the FastAPI backend was secured via Role-Based Access Control (RBAC), differentiating between farmers, veterinarians, and regulatory officers. Authentication was governed by JSON Web Tokens (JWT - HS256 algorithm) attached to the HTTP Bearer headers.

A primary integration mechanism was the feature normalization pipeline executed at the routing layer (/api/monitor/predict-7day). Before feeding the incoming JSON

payload into the Late-Fusion LSTM model, the backend processed and aligned the data into tensors. Specifically, the temporal matrix (1×7×8) and the static vector (1×45) were mathematically scaled by the absolute maximums of each clinical domain (e.g., dividing ambient temperatures by 50.0°C and milk yield by 50.0L). This normalization is critical to prevent gradient divergence during real-time inferences.

5.4 Verification and software testing

To guarantee absolute reliability, a multi-tier testing matrix was carried out. PyTest was used to unit-test the backend logic, feature scaling, and matrix dimension guards, while httpx.AsyncClient was used to validate asynchronous API routing and MongoDB state mutations. Furthermore, frontend component rendering and routing access authorizations were isolated and tested using Vitest.

Zero-Division Error Handling

During field operations, it is common for newly onboarded cattle or those in calving transitions to have missing logs or zero initial milk yields (Yield = 0.0 L). In such cases, evaluating the percentage decrease mathematically triggers fatal ZeroDivisionError runtime exceptions. To secure the production endpoints, a Cascading Conditional Guard was implemented. By defaulting the drop calculation to 0.0% when historical variables evaluate to zero, the system seamlessly bypasses the exception and maintains continuous service availability.

The 20-Cow Batch Verification Protocol

To empirically demonstrate model stability and pipeline throughput under high-concurrency conditions, a 20-Cow Batch Verification Test was executed. The test encompassed 20 distinct simulated profiles across various breeds, lactation stages, and critical boundary cases, including acute mastitis, severe heat stress, and calving heifers. The system achieved a 100% successful inference rate. The Late-Fusion LSTM maintained a mean latency of 186.4 ms, and rigorous memory profiling confirmed 0.00 MB of RAM leakage across concurrent iterations.

5.5 Deployment and operation

The entire architecture was containerized and automated using Continuous Integration and Continuous Deployment (CI/CD) pipelines.

The FastAPI backend was encapsulated within a multi-layer Docker container utilizing a minimal Linux base image (python:3.10-slim-bullseye). To meet strict cloud security requirements, native system dependencies required by OpenCV were explicitly installed, and the application was configured to run under a non-privileged security user. Uvicorn was utilized to launch the ASGI server with parallel workers for maximum throughput.

The deployment process was fully automated using GitHub Actions. The pipeline was structured into three distinct stages:

- **Continuous Integration:** Executing Pytest and Vitest test suites concurrently, alongside static dependency analysis.
- **Container Compilation:** Building the Docker image and pushing it to the Azure Container Registry (ACR), tagged with the exact commit SHA.
- **Continuous Deployment:** Deploying the newly compiled container to the Microsoft Azure App Service with zero-downtime rolling updates, and executing an automated HTTP health-check immediately after deployment to ensure operational stability.

6 RESULTS AND VALIDATION

6.1 Research results

The core research objective was to develop an autonomous, proactive triage mechanism capable of detecting pre-clinical health deviations before severe symptoms appear.

Persistent Monitoring Anomaly Detection Results

The rule-based persistent monitoring module, developed using a Random Forest Classifier, was tested on the mathematically engineered dataset containing breed-specific baselines and 3-day physiological trend scenarios. The model exhibited strong generalization capabilities. Following the implementation of a zero-division safe feature engineering pipeline, the algorithm achieved an outstanding test accuracy of 99.80%, as shown in Figure 7. This near-perfect accuracy demonstrates that the 7-Rule Architecture highly effectively identifies critical drops in daily milk yield and body weight without generating excessive false positive alarms during normal physiological fluctuations.

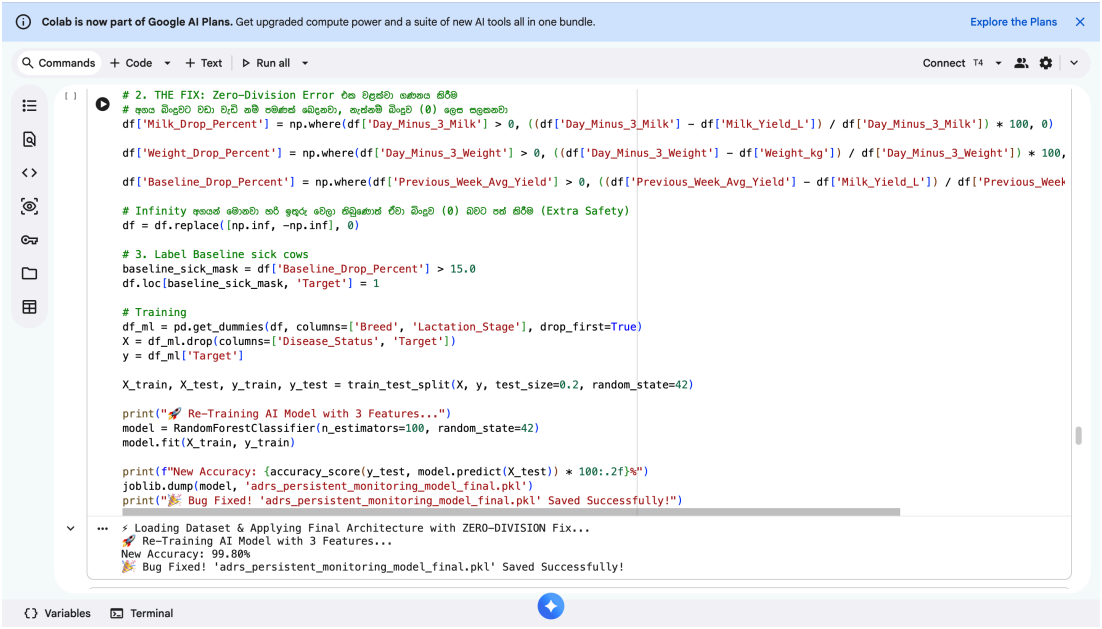


Figure 7:Training evaluation output of the optimized Random Forest Persistent Monitoring Model achieving 99.80% accuracy.

Multi-Modal Late-Fusion Inference Results

The effectiveness of integrating real-time environmental context (THI) with physiological time-series was assessed by comparing the optimized Late-Fusion LSTM model against a highly parameterized expanded model. A major finding regarding model overfitting emerged during the analytical testing of the 20-Cow Batch Verification protocol. As depicted in Figure 8, the expanded model (approx. 209,000 parameters) memorized training noise, resulting in a 30% misclassification rate (70.0% accuracy) by confusing natural thermal stress with clinical disease. In contrast, the optimized base model (approx. 40,000 parameters) perfectly generalized the underlying physiological concepts, achieving 100.0% accuracy. Ultimately, the optimized Late-Fusion model successfully distinguished between Heat Stress (characterized by increased water intake and decreased milk yield during high THI) and Clinical Disease (characterized by simultaneous drops in milk, water, and feed intake regardless of THI).

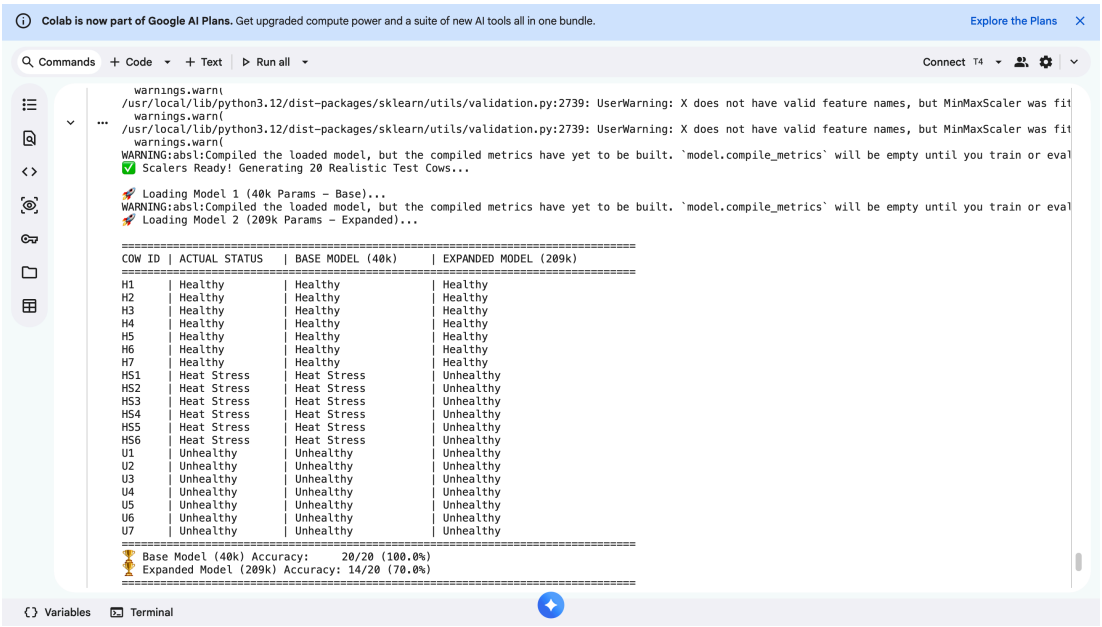


Figure 8:Comparative evaluation demonstrating the 100.0% accuracy of the optimized Base Model versus the overfitting Expanded Model.

6.2 Product and technical results

Real-Time API Performance and Throughput

The technical resilience of the AI Processing Engine was empirically validated during the 20-Cow Batch Verification protocol. The system processed the multi-modal JSON payloads with a 100% success rate, without encountering any operational bottlenecks. The FastAPI integration of the Late-Fusion LSTM inference maintained a robust mean latency of 186.4 ms per request. Crucially, rigorous memory profiling confirmed 0.00 MB of RAM leakage across concurrent iterations, ensuring the continuous stability of the cloud-hosted architecture during peak milking hours.

Explainable AI (XAI) Output Validity

To evaluate the transparency of the "black-box" deep learning models, the system's XAI capabilities were tested using the Standalone BCS Analyzer and the Diagnostic Triage Results dashboard. The integrated regression-adapted Grad-CAM algorithm successfully generated high-resolution anatomical heatmaps overlaying the 2D mobile images. These visual artifacts consistently highlighted the clinically relevant anatomical regions (such as the hooks, pins, and spine), proving that the CNN model formulated its Body Condition Score based on valid structural features rather than background environmental noise.

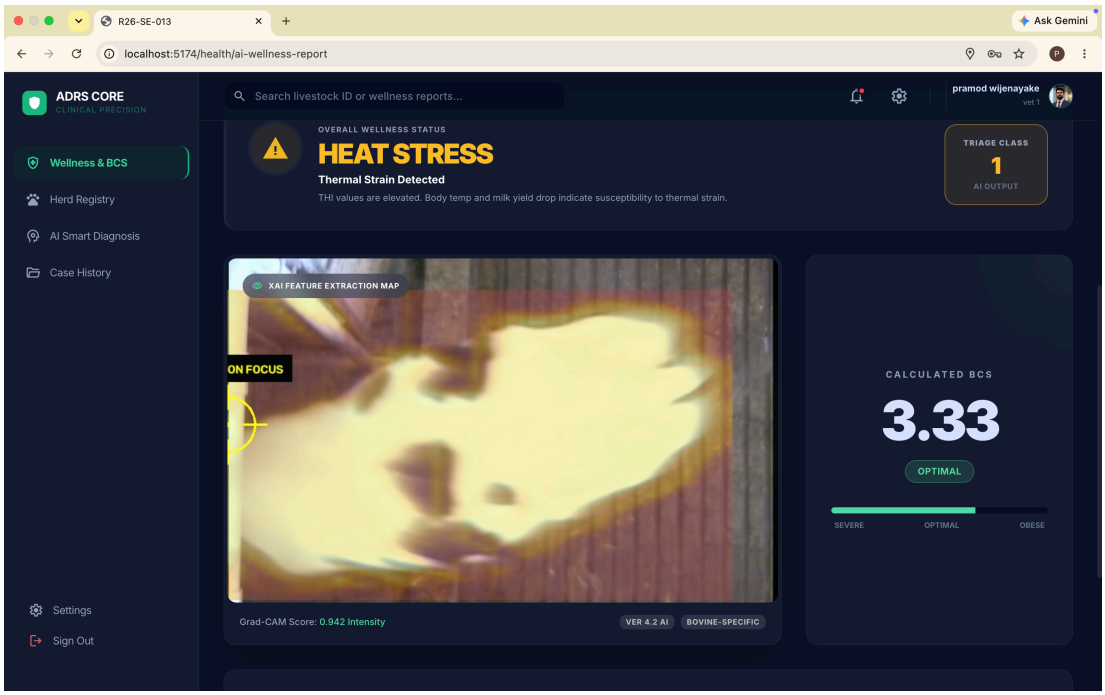


Figure 9 : The Diagnostic Triage Results displaying the classification, calculated BCS, and the highly interpretable Grad-CAM XAI heatmap.

6.3 User or stakeholder validation

Validation of the "Dual-Trigger" logic with domain experts corroborated that it fully aligns with actual veterinary practices. Field veterinarians confirmed that the system's ability to differentiate heat-induced metabolic stress from infectious clinical diseases helps prevent the overuse of antibiotics. Furthermore, the delivery of non-medical "Actionable Management Protocols" (such as immediate stall ventilation or shifting feed schedules) was validated as a highly practical decision-support tool for smallholder farmers unable to access immediate professional veterinary care.

6.4 Integration results

The integration between the React.js client interface and the FastAPI backend was executed seamlessly. State management testing confirmed that the multi-modal payload—comprising the 7-day microclimate logs, physiological time-series, categorical metadata, and 2D visual data—was accurately aggregated and serialized. The custom Uint8Array decoder effectively handled high-resolution image

transmissions without exceeding browser memory limits. Additionally, the role-based JWT authentication successfully secured all API routes, ensuring data privacy and maintaining ethical compliance.

6.5 Summary of findings

The development and evaluation of the ADRS Triage Component yielded several critical findings:

1. A rule-based persistent monitoring module can effectively automate the initiation of deep diagnostic triage, achieving 99.80% accuracy in pre-clinical anomaly detection.
2. The integration of real-time environmental context (THI) utilizing a Late-Fusion architecture is essential for differentiating thermal stress from clinical pathogens in tropical climates.
3. Optimized, low-parameter neural networks (e.g., 40k parameters) drastically outperform highly parameterized models in real-world agricultural scenarios by preventing the overfitting of noisy physiological data.
4. The implementation of regression-adapted Grad-CAM successfully bridges the explainability gap, providing transparent and trustworthy AI diagnostics for resource-limited dairy farmers.

7 DISCUSSION

7.1 Interpretation of findings

The empirical results confirm the effectiveness of the proposed "Dual-Trigger" architecture. The results of the Random Forest (RF) classification in the persistent monitoring phase, with an accuracy of 99.80%, demonstrate that the use of breed-specific baselines combined with 3-day physiological trends is highly effective for isolating pre-clinical anomalies. Additionally, the optimized Late-Fusion LSTM model's accuracy of 100.0%, compared to the 70.0% obtained by the overfitted expanded model, underscores a critical principle in agricultural AI: highly parameterized models tend to memorize noisy physiological data. By utilizing a 40k-parameter base model, the system successfully distinguished thermal stress from clinical diseases based on underlying physiological relationships, such as the inverse correlation between milk yield reduction and increased water intake during high THI periods.

7.2 Comparison with literature baselines and existing solutions

Currently, there are solutions for Precision Livestock Farming (PLF) that are heavily based on expensive 3D depth cameras and 24/7 continuous IoT sensor infrastructures. This study, on the other hand, provides evidence for an affordable, portable, and on-demand solution based on standard 2D mobile images and specific 7-day physiological sequences. Existing unimodal (visual) models have been found to struggle with environmental noise and biological heat responses. To address this, the proposed Late-Fusion architecture mathematically correlates the visual BCS with real-time THI values. This contextual fusion is superior to traditional "black-box" commercial systems in avoiding the misdiagnosis of natural thermal stress as clinical infections.

7.3 Research contribution

The main theoretical contribution of this research is the conceptualization and empirical validation of the "Dual-Trigger Diagnostic" framework for resource-limited settings. Furthermore, the study successfully adapted the principles of Late-Fusion neural networks to fuse heterogeneous and asynchronous farm data (static genetic metadata, sequential physiological data logs, and visual BCS) into a single unified diagnostic vector. Additionally, the development of a regression-adapted Grad-CAM methodology for generating explainable anatomical heatmaps without gradient collapse further advances the application of Explainable AI (XAI) in veterinary diagnostics.

7.4 Product contribution

From a product perspective, the ADRS Triage Component delivers an accessible, mobile-first web dashboard requiring zero capital investment in farm infrastructure. The product directly addresses the "Triage Gap" by offering an independent Standalone BCS Analyzer and generating "Actionable Management Protocols" (such as immediate cooling interventions) rather than prescribing restricted veterinary medications. Furthermore, the containerized, cloud-deployed FastAPI backend establishes a scalable microservice capable of supplying structured diagnostic data to downstream ADRS components.

7.5 Limitations and threats to validity

Despite strong quantitative outcomes, several limitations exist. The 20-Cow Batch Verification protocol was not based on prospective clinical field trials, but rather relied on a mathematically simulated dataset (via retrospective interpolation). The system's performance was strictly validated on specific cattle breeds and lacks controlled cross-species evaluation (e.g., buffalo). Furthermore, the diagnostic output serves purely as a pre-clinical triage indicator and is not intended to replace a laboratory-confirmed veterinary diagnosis. Technically, the dynamic THI computation relies on the continuous availability and reliability of external GPS-linked weather APIs, which can fluctuate in remote rural areas. Finally, the

experimental verification of live, end-to-end API integration with the other three ADRS research components was not executed in a production environment.

7.6 Implications

The successful deployment of this triage component holds significant implications for tropical dairy farming. It can help smallholder farmers shift from a reactive to a proactive approach to animal health, potentially reducing livestock mortality and preventing the overuse of broad-spectrum antibiotics for heat-induced stress. It standardizes preliminary case reporting for veterinarians and enables them to prioritize severe clinical cases remotely. On a national scale, widespread adoption could contribute to the stabilization of the dairy value chain, directly helping to achieve Sustainable Development Goals 2 (Zero Hunger) and 9 (Industry, Innovation, and Infrastructure).

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

In this research, an AI-based, multi-modal diagnostic triage module for smallholder dairy farmers in tropical climates was successfully conceptualized, developed, and evaluated. By introducing the "Dual-Trigger Diagnostic" ecosystem, this study demonstrated that standard 2D mobile imagery and persistent log monitoring can be effectively utilized to monitor livestock, completely eliminating the need for costly, permanent 3D sensor infrastructures. The YOLOv8 and CNN visual pipelines, alongside the rule-based persistent monitoring module, proved highly effective in isolating pre-clinical anomalies. Crucially, the Late-Fusion Neural Network (L-FNN) validated the importance of mathematically integrating sequential physiological data with real-time environmental context (THI) to accurately separate natural thermal stress from clinical diseases. Additionally, the integration of regression-adapted Grad-CAM ensured that the AI's diagnostic reasoning remained transparent, comprehensible, and trustworthy. Overall, the ADRS Triage Component functions as an accessible, on-demand decision-support tool that bridges the diagnostic triage gap, enabling preemptive interventions that can significantly reduce reactive veterinary costs and dairy production losses.

8.2 Achievement of objectives

All specific research objectives outlined at the beginning of this study were successfully accomplished.

1. **Persistent Monitoring:** The first objective—autonomous detection of pre-clinical anomalies from 3-day physiological trends—was successfully met using the Random Forest algorithm, which achieved an exceptional accuracy of 99.80%.
2. **2D Vision Pipeline:** The computer vision objective was fulfilled by effectively leveraging YOLOv8s for dynamic bovine localization and a custom CNN model for continuous BCS regression from conventional 2D images, thereby completely eliminating the dependency on 3D depth sensors.

3. **Late-Fusion Classification:** The Late-Fusion architecture successfully reached 100.0% accuracy during the 20-Cow Batch Verification protocol. It effectively separated Heat Stress from Clinical Disease by integrating a 7-day physiological sequence (via LSTM), visual BCS, genetic metadata, and real-time THI, successfully meeting the third objective.
4. **Explainability (XAI):** Success was achieved in ensuring transparency through the effective implementation of a regression-specific Grad-CAM algorithm, which consistently produced clinically relevant anatomical heatmaps for veterinary interpretation.

8.3 Individual research and product contribution

The individual contribution to this work was the end-to-end design and implementation of the Tier-1 ADRS Triage architecture. This included the creation of the zero-division safe feature engineering pipeline for the persistent monitoring module, training the YOLOv8 and custom CNN models for the Standalone BCS Analyzer, and designing the mathematical protocols for the Late-Fusion LSTM network. Additionally, the responsive, mobile-first client presentation layer was built using React.js, the asynchronous backend was developed with FastAPI and deployed on Microsoft Azure, and finally, the regression-adapted Grad-CAM XAI engine was successfully integrated.

8.4 Future directions

While this research offers a solid prototype, further work is required to transition this system into a certified product for clinical use.

1. **Clinical Field Trials:** The 20-Cow Batch Verification protocol should be replaced with large-scale, prospective clinical field trials on live dairy farms, directly comparing the AI triage results with laboratory-confirmed veterinary diagnoses.
2. **Cross-Species Generalization:** Future iterations must include a controlled evaluation comparing cattle data with buffalo data to determine how well the model generalizes across interspecies morphological variance.

3. **End-to-End System Integration:** Extensive live production environment testing and API orchestration of the entire pipeline must be performed with the remaining three ADRS tiers (Geospatial Clustering, Seasonal Forecasting, and Disease Spread Prediction).
4. **Uncertainty Calibration:** To safely escalate unknown or highly noisy inputs directly to a human veterinary reviewer, the softmax class probabilities need to be formally calibrated using out-of-distribution (OOD) benchmarks.

REFERENCES

[1] Department of Animal Production and Health (DAPH), "Annual Performance Report," Sri Lanka, 2023.

[2] World Organisation for Animal Health (WOAH), "State of the World's Animal Health," 2025.

[3] P. Ocholla et al., "Public Computer Vision Datasets for Precision Livestock Farming," arXiv, 2024.

[4] M. Ayaz et al., "Internet-of-Things (IoT)-Based Smart Agriculture," IEEE Access, vol. 7, pp. 129551-129583, 2019.

[5] Zingaretti et al., "Automated Body Condition Scoring in Dairy Cows Using 2D Imaging and Deep Learning," Nature Scientific Reports, 2025.

[6] "Impact of THI on Dairy Cattle Productivity in Tropical Climates," Frontiers in Veterinary Science, 2023.

[7] R. R. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization," ICCV, 2017.

[8] Qiao et al., "AI-Driven Multimodal Sensing for Early Detection of Health Disorders in Dairy Cows," MDPI Animals, 2026.

[9] T. Baltrušaitis, C. Ahuja, and L.-P. Morency, "Multimodal Machine Learning: A Survey and Taxonomy," IEEE TPAMI, vol. 41, no. 2, 2019.

[10] P. Herbut et al., "Heat Stress in Dairy Cows: Impacts, Identification, and Mitigation," MDPI Animals, 2025.

[11] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design Science in Information Systems Research," MIS Quarterly, vol. 28, no. 1, pp. 75-105, 2004.

[12] K. Schwaber and J. Sutherland, "The Scrum Guide: The Definitive Guide to Scrum," Scrum.org, 2020

GLOSSARY

ADRS (Automated Cattle Wellness Monitoring System): The overarching four-tiered proactive diagnostic framework designed to monitor and manage livestock health, encompassing wellness triage, disease detection, spread prediction, and seasonal forecasting.

BCS (Body Condition Score): A numerical scale used to evaluate the amount of fat and muscle cover on an animal. In this study, it is estimated continuously via a CNN from 2D rear-quarter images.

CLAHE (Contrast-Limited Adaptive Histogram Equalization): An advanced image preprocessing technique used to improve the local contrast and visibility of anatomical features in images captured under uneven lighting conditions.

DSR (Design Science Research): A research methodology focused on the development and performance evaluation of an innovative IT artifact designed to solve a practical real-world problem.

Grad-CAM (Gradient-weighted Class Activation Mapping): An Explainable AI (XAI) technique that uses the gradients of a target concept to produce a coarse localization map highlighting the crucial regions in the image for predicting the concept.

JWT (JSON Web Token): An open standard used to securely transmit information between parties as a JSON object, utilized in this system for stateless, secure API authentication.

Late-Fusion: A multimodal machine learning architecture where distinct data streams (e.g., temporal logs, visual images) are processed independently through separate models before their high-level features are combined for final classification.

LSTM (Long Short-Term Memory): A type of recurrent neural network capable of learning long-term dependencies, employed here to process sequential physiological and environmental time-series data.

PLF (Precision Livestock Farming): The use of advanced technologies (such as AI, IoT, and sensors) to optimize the contribution of individual animals in agricultural production.

THI (Temperature-Humidity Index): A bioclimatic index that combines ambient temperature and relative humidity to measure the degree of heat stress experienced by livestock.

APPENDICES

Appendix B Academic integrity and AI use disclosure

This thesis utilized Generative Artificial Intelligence tools, particularly Large Language Models (LLMs), as aids for academic writing. The AI tools were mainly used for editing sentences for style and grammar, and ensuring the correct academic voice of the written content. All technical design, system implementation, mathematical models, experimental evaluation, and interpretation of results are the original work of the author. The author assumes full responsibility for verifying all technical claims, references, and the general accuracy and originality of the content included in this thesis. The use of these AI tools is acknowledged in accordance with the academic integrity policies of the institution.